

Phase 2: Information Architecture, Wireframing & User Flow Design

CMPS 4200 HCI: AI-Enhanced UI/UX | Fall 2026

Duration	Weight	Presentations
2 weeks (weeks of Sep 21 and Sep 28)	20% of final grade	In class, week of Oct 5

Objective

Design the structure and flow of your application, with particular attention to how users will interact with AI features. Focus on creating intuitive pathways that make AI capabilities accessible and trustworthy.

Good AI interface design depends on how your model actually behaves: how long it takes, how it fails, and how reliable its output is. This phase therefore includes a short **AI technical spike** so that your loading, error, and trust designs are based on real model behavior rather than assumptions. It is also your chance to confirm that the model you selected in Phase 1 works for your use case while there is still time to change it.

Key Dates

When	What happens
Week of Sep 21	Phase 2 kickoff and studio time. Begin site map, flows, and the technical spike.
Week of Sep 28 (in class)	AI technical spike due. Bring draft flows and wire-frames for team-to-team critique; each team gives and receives structured feedback.
Week of Oct 5	All deliverables due by the end of the week. Phase 2 presentations in class.

Grade Breakdown

Component	Weight
1. Information Architecture	20%
2. User Flow Diagrams	30%
3. Low-Fidelity Wire-frames	30%
4. AI Technical Spike	10%
5. Presentation	10%

1. Information Architecture (20%)

- **Site Map:** A hierarchical structure of all pages/screens.
- **Content Strategy:** Define what information is displayed where.
- **AI Integration Points:** Clearly mark where and how AI models are integrated. For each point, name the model, its input, its output, and the expected response time (taken from your technical spike).
- **Data Flow Diagrams:** Show how data moves from user input through the front-end, back-end, and AI service, and back to the user. Indicate where data is stored and which user data is sent to a third-party AI provider.

2. User Flow Diagrams (30%)

- **Primary User Flows:** Design flows for 3-5 key user tasks. Label each flow with the Phase 1 persona it serves.
- **AI Interaction Flows:** Detailed flows showing:
 - How users provide input to AI models
 - Loading and processing states
 - How AI results are presented
 - Error handling and fallback scenarios
 - How users edit, reject, regenerate, or undo AI results
- **Edge Case Scenarios:** Document at least three, including (a) the AI service is unavailable, times out, or hits a rate limit; (b) the AI returns a poor, wrong, or empty result; and (c) the user provides invalid or unsupported input. Consider also what happens if the output is biased or inappropriate.

AI-Specific Flow Considerations

- **Loading States:** How do users understand that the AI is processing? Base this on the latency you measured. As a rule of thumb, responses under about 1 second need little feedback; 1-10 seconds need a clear progress message; longer waits need progress detail, streaming, or a way to cancel.
- **Error Recovery:** What happens when AI models are unavailable or produce poor results? Users should never reach a dead end.
- **User Control:** How can users modify, reject, or refine AI suggestions?

3. Low-Fidelity Wire-frames (30%)

- **Complete Wire-frame Set:** Cover every screen in your site map and all major user states.
- **AI Interface Elements:** Design components for:
 - AI input interfaces (voice, text, image upload, etc.)
 - AI processing feedback (loading indicators, progress bars)
 - AI result presentation (confidence information, alternative suggestions)
 - User control (edit, reject, regenerate, undo)
- **Required AI States:** For each AI feature, wire-frame all five states: *empty/initial*, *loading*, *success*, *low-confidence or uncertain result*, and *error*.
- **Annotations:** Label each AI component with the flow and state it belongs to.

A note on confidence scores. If your model produces meaningful scores (for example, a classifier), show them in plain language that users can act on. Large language models and other generative models do not produce reliable confidence numbers, so do not invent percentages. Use alternatives instead: cite sources or evidence, give a short rationale, offer several alternatives, use hedged wording, or prompt users to verify important results.

Tools: Figma, Sketch, Adobe XD, or similar.

4. AI Technical Spike (10%)

A small, working piece of code that tests your chosen model(s) before you design around them. This is not part of your final application and does not need a user interface.

- **Working Call:** A script or notebook that sends at least 5 realistic sample inputs from your use case to your chosen model(s) and records the outputs.
- **Latency:** Record the response time for each call and report the minimum, average, and maximum.
- **Failure Testing:** Trigger at least 2 failure conditions (for example, malformed input, an invalid key, a timeout, a rate limit, or an empty result) and record exactly what the model returns.

- **One-Page Summary:** Report latency, output quality, failure behavior, rate limits and cost, and how these findings shaped your loading and error designs. If the model is not suitable, you may switch models; justify the change.

5. Presentation (10%)

Each team has 10 **minutes plus 3-5 minutes of Q&A**, strictly timed. Every team member should speak.

What to Submit

- Presentation slides.
- A team contribution statement: a short list of what each member contributed, agreed to by all members. This may be used to adjust individual grades.